Enhancing Content-Based Music Recommendation Systems for Middle Eastern Music

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Abstract

This paper explores the potential of Wikipedia as a source of detailed music metadata for enhancing music recommendation systems, particularly addressing the gap in metadata representation for regions like the Middle East. Through the implementation of a sophisticated methodology for collecting and cleaning Wikipedia-sourced metadata, the study aims to overcome the limitations of collaborative filtering techniques prevalent in platforms such as Spotify, YouTube and Apple Music. The analysis reveals a significant improvement in metadata quality, with the cleaned Wikipedia dataset for Iran and India showing a significant increase in metadata richness compared to the publicly available Musicbrainz dataset, underscoring the potential of Wikipedia to enhance content-based recommendation systems.

The research provides a significant contribution to the music recommendation field by outlining a methodical approach to enhancing metadata accuracy and richness, showcasing the benefits of utilising open-source Wikipedia information over existing datasets for improving recommendation systems. My findings underscore the importance of comprehensive metadata in delivering more accurate and diverse music recommendations and sets a new precedent for leveraging untapped open-source resources to advance music recommendation technologies.

Research Ethics Approval

This project was planned in accordance with the Informatics Research Ethics policy. It did not involve any aspects that required approval from the Informatics Research Ethics committee.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

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Chapter 1

Introduction

Music recommendation systems are pivotal in the way individuals are able to discover and interact with music today. These systems, utilised by platforms like Spotify, YouTube and Apple Music, utilise sophisticated algorithms to suggest songs, artists and genres to users, enhancing their listening experience. Central to the effectiveness of these recommendation systems is the use of musical metadata, which includes detailed information about the tracks, such as composer and lyricist information.

Music recommendation systems have evolved significantly over the years, starting from simple algorithms that would base their recommendations on genre classification, to complex systems employing intricate collaborative filtering and machine learning techniques. Collaborative filtering, in particular, plays a crucial role in modern recommendation systems. It analyses the similarities between users and their individual behaviours to predict the music that they would most likely enjoy next. Whilst platforms like Spotify and YouTube have developed powerful approaches through this technology, they often face challenges such as the cold start problem, where new songs or artists with limited user interaction data are less likely to be recommended [Schedl, 2019, Weng et al., 2022].

To mitigate the issues inherent in collaborative filtering, content-based recommendation systems can be presented as a potent solution. This type of system analyses the content of the music itself, which allows for recommendations to be made based on the intrinsic properties of the songs, rather than utilising user interaction patterns. This approach significantly benefits from having rich metadata available for the music, enabling the identification of similar traits between tracks that go beyond user interaction [He et al., 2017].

However, the effectiveness of content-based recommendation systems is reliant on the availability and richness of metadata. This is particularly challenging for music from regions like the Middle East, where the datasets often lack comprehensive metadata. The absence of detailed composer and lyricist information not only dampen the effectiveness and accuracy of their recommendations, but also undermine the representation of the cultural heritage these musical works hold.

While Spotify, YouTube and Apple Music excel in providing personalised music experiences, their reliance on collaborative filtering underscores a significant dependence on user interaction data. This approach will often bias recommendations towards well-known artists and songs, effectively sidelining less popular music that might have aligned with the user's interest. Content based recommendation systems offer a solution to this, by focusing on the music's content, albeit at a cost of requiring detailed metadata to be present in the datasets [Knees and Ferraro, 2022, Santana and Domingues, 2022].

In response to these challenges, this work will explore the potential of Wikipedia as a source of detailed music metadata, focusing on composer and lyricist information. Wikipedia's extensive archive of information offer an untapped resource for enriching content-based recommendation systems, especially for music genres and regions like the middle east which are currently underrepresented in the field of metadata in existing datasets.

A major challenge in sourcing metadata for Middle Eastern Music from Wikipedia stems from its scattered nature on the platform, made even further problematic by the absence of joint efforts made to aggregate and clean the data into a usable format. Indeed, the information regarding musical metadata we are interested in is usually contained within an artist's discography page. Due to the nature of Wikipedia, this data will have been submitted by a user of the system, often scattered across the page and unreliably written in a format that can be different going from one page to the next. This dispersion of the data requires a system that implements a sophisticated approach to data scraping and cleaning to ensure only useful information is stored in our database, that goes beyond simple extraction of the information. Whilst a single researcher could take on the task of individually parsing each artist's discography by hand to collect the metadata, this would be a long and cumbersome task that would not address the issue of building a dataset covering the entirety of Middle Eastern music in a timely manner [Latif et al., 2019, Rico et al., 2021].

To harness Wikipedia's potential, a comprehensive approach must be taken to create a cohesive and complete dataset that collects and cleans the different formats the data is currently stored in, into a single, unified structure. To address the problem of scattered data sources, I have created a method to systematically identify and retrieve relevant Wikipedia pages, carefully scrape and clean the requested data, and create a reproducible system that could perform this process over any number of countries.

This method involves a multi-layered process that begins with the development of a system to carefully identify the correct pages that host the data we need. We then identify the patterns shared across these pages and by utilizing complex data scraping techniques we collect and store all the metadata from each artist [Akbaş, 2023, Yaghmour et al., 2021].

Following this data extraction phase, the next important step is to clean the collected data. This involves not only organizing the data and cleaning the textual discrepancies from song to song but also specifically solving differences in spelling that might lead a system using the database to incorrectly identify instances of the same artist as being different artists. This issue is extremely prevalent due to the nature of data on Wikipedia coming from multiple sources that can result in many variations of the same artist being inputted on various different pages, even though all instances represent the same artist. Through this method, my system is equipped to analyze the metadata collected and clean it with a high level of precision, which facilitates more accurate music recommendations to the end users. This approach not only solves the problems presented by the current state of the Middle Eastern metadata being scattered across the internet but also sets a new standard for how musical applications will be able to utilize open-source information to further enrich the user experience [Epure and Hennequin, 2023].

This work contributes to the field of music recommendation systems and metadata analysis in several significant ways. The key contributions of this paper are outlined as follows:

- **Literature Review:** I will provide an extensive review of the existing literature on music recommendation systems, with a focus on the evolution from collaborative filtering to content-based recommendation systems. This review will highlight the methodologies that I will utilise when developing the different architectures of my project.
- **Data Collection System:** I will detail the development of a sophisticated system for the systematic identification, retrieval and initial processing of music metadata from Wikipedia. This will include the practises used for navigating the dispersed data used to create a preliminary dataset.
- **Data Cleaning System:** My work will further extend to the design and execution of a comprehensive data cleaning framework. This system will be tailored to addressing the challenges faced with user-sourced data, specifically regarding it's reliability for use in a content-based recommendation system.
- **Content-Based Recommendation System:** Leveraging the cleaned dataset, I will implement a content-based music recommendation system. This system will leverage the detailed metadata that has been collected and cleaned, aiming to address the limitations of collaborative filtering. By focusing on the musical attributes of each song, it will create more accurate recommendations.
- **Web Platform:** To facilitate interaction with the dataset and recommendation system, I will present an architecture of a user friendly platform. This platform includes both backend and frontend components, enabling data insights and visualizations that highlight the richness of metadata and it's impact on music discovery.
- **Experimental Results:** Through rigorous testing and evaluation, I will provide empirical evidence to demonstrate the effectiveness of my content-based recommendation system compared to traditional methods. My experimental results will showcase the improvements made to the dataset using my cleaning framework, and compare the density and quantity of metadata within each of the datasets.
- **Conclusion:** Building upon the development of my system and experimental results, I will critically assess the broader impact of my work on the accessibility and understanding of middle eastern music metadata. This will include evaluating the implications of my research on the current scope of musical datasets, and

commenting on future work that could expand the potential my project has to reach a global scale.

By addressing each of these critical aspects, this paper aims to advance the field of music recommendation systems by highlighting the importance of comprehensive and accurate musical metadata. My methodologies and findings will underscore the potential of leveraging open source platforms like Wikipedia to enrich music metadata, thereby enhancing the music discovery experience for users worldwide.

Chapter 2

Literature Review

This chapter comprehensively approaches the existing literature surrounding the methodologies I will implement. Central to digital music applications, music datasets are indispensable for a wide array of uses, from serving as the backbone of recommendation systems to supporting academic research. They offer a comprehensive view of musical compositions that extends beyond basic classifications, like genre and album information, to encompass an intricate web of creators that are behind the music. Despite their important role, prevailing datasets exhibit serious shortcomings, particularly in their coverage of middle eastern music, underscoring a gap in detailed metadata including composer and lyricist information.

I will then discuss the significance of robust data collection methods, highlighting web scraping as a pivotal technique for metadata extraction in my methodologies. The specific intricacies of web scraping are discussed, alongside potential legal and technical challenges that surround them, leading into the crucial phase of data cleaning that will standardise the dataset and mitigate any inconsistencies.

This leads into discussing how content-based recommendation systems can leverage these cleaned datasets, and particularly the rich metadata they contain, to generate reliable recommendations.

This chapter lays the groundwork for understanding the methodologies I have utilised throughout the development of my project and will not only shed light on the technical and ethical considerations involved in the creation of my datasets, but also underscores the critical role they play in advancing digital music applications and research.

2.1 Existing Music Datasets and their Limitations

2.1.1 Overview of Existing Music Datasets

For digital music applications, having comprehensive and accessible datasets is of extreme importance. They serve as the foundation for various use-cases, ranging from recommendation systems to academic research, allowing each musical composition's information to be broken down to reveal the various attributes that identify it. This data not only includes which albums and genres the songs belong to, but can also display deeper connections showing the composers and lyricists who's artistic vision was behind its creation.

A major open access database stands at the top in regards to their unique contributions to their efforts of aggregating music data. Musicbrainz, which has amassed a vast catalog of music releases and their associated metadata [Schindler et al., 2012].

2.1.2 The Current Limitations of MusicBrainz

Despite MusicBrainz's invaluable contributions to the space as a whole, their system's show serious limitations when it comes to the scope of their datasets, specifically in regards to Middle Eastern specific music. Indeed, while both platforms may offer basic data like track names and artist information, they frequently lack detailed metadata about composer and lyricist information. This gap not only negatively affects the completeness of their datasets but also limits their usability for both specialised research and applications that would otherwise have utilised this data to create personalised experiences for their users [Bertin-Mahieux et al., 2011].

2.1.3 Past Attempts to Solve Metadata Incompleteness

The Million Song Dataset project, created by Bertin-Mahieux in 2011, highlights the challenges of aggregating an extensive music metadata dataset. This ambitious project was aimed to provide a large-scale, comprehensive dataset for the development of music systems. Whilst it set a foundational standard for attempting this task of consolidating music metadata, it also highlighted the complex issues including ensuring the representation of composer and lyricist information details were accurate.

This project was predominantly focused on Western music, with very little focus on any other regions, further highlighting the gaps in global music metadata coverage, which is especially prevalent in the Middle East.

2.1.4 Addressing the Gap in Middle Eastern Metadata

These challenges highlight the critical need for a targeted effort to bridge the gaps in metadata found in music datasets, especially for underrepresented regions like the Middle East. The Limitations of existing datasets not only restrict the potential scope of research initiatives, but also the effectiveness of recommendation systems built upon these datasets.

2.2 Data Collection

2.2.1 The Importance of Data Collection in Music Research

The process of data collection is a vital aspect of research, especially in the field of music technology where large volumes of information are scattered across the web.

Web Scraping stands out as a critical technique used in extracting various data from online sources.

This technique specializes in gathering vast amounts of unstructured data available on the internet, and processing it into an analyzable format that will be suitable for further data cleaning and insight generation steps [Tripathi and Nema, 2019, Aljanaki et al., 2013]. When planning to compile large datasets of music metadata, web scraping is a significant step.

2.2.2 Web Scraping as a Technique for Metadata Extraction

Web scraping uses sophisticated algorithms to automatically navigate the HTML structure of web pages, and extract the useful data, bypassing the traditional, manual methods of data collection which are not only extremely time-consuming but also prone to man-made errors.

This approach allows researchers and developers to create large datasets efficiently and accurately. For a project like creating a music database with comprehensive metadata, web scraping becomes an essential tool that enables the collection of data from multiple web pages dispersed across the internet.

2.2.3 Challenges Faced in Web Scraping

One of the major problems with web scraping is handling the many variations that web pages can present. Websites can vary widely in their formatting and this is especially the case with Wikipedia where any two different artist pages could have completely different authors. This requires a web scraper to be highly adaptable and capable of understanding various patterns present, both in the wording and formatting from page to page.

Indeed, the legal considerations surrounding web scraping practices should not be ignored and whilst Wikipedia is open to scraping, the specific rules and regulations of a website should be respected when building a scraper [Gold and Latonero, 2018].

2.2.4 Processing and Preparing the Scraped Data

Another significant aspect of web scraping is the further cleaning and data processing that is performed on the scraped data. The data obtained from web scraping is often cluttered with irrelevant information which requires extensive cleaning to result in the data being usable.

This issue becomes even more pronounced when considering data scraped from Wikipedia. On Wikipedia, there is no standardized format for storing musical data, leading to significant variations in how information is presented across pages. Formats range from tables to lists, accompanied by inconsistencies in headings. Additionally, spelling errors, whether accidental or resulting from variations of the same name, further complicate data extraction. All of these errors need to be accounted for when data scraping and require sophisticated algorithms capable of standardizing the metadata, and removing the excess waste [Luscombe et al., 2021, Salvador et al., 2023].

2.3 Data Cleaning

2.3.1 Addressing the Data Inconsistencies

In the process of building a dataset for an application such as a content-based recommendation system, data cleaning is a complex yet essential task. At this stage in the data preparation, we take the raw and disorganised data collected through methods like web scraping, and we transform it into a cleaned, reliable dataset. This involves a range of advanced techniques such as refining, normalising and encoding the data to ensure the dataset is usable [Thorat et al., 2023, Kumar and Kumar, 2022].

One of the initial steps taken during data cleaning is the identification and removal of inaccuracies and inconsistencies. This not only involves handling duplicate entries, but also all kinds of typing errors and irrelevant information. This is essential to maintaining the usability of the dataset and ensures, in the context of music metadata, that any textual errors in artist names or song titles due to spelling mistakes or transliteration errors are handled correctly.

2.3.2 Normalisation of Data for Uniformity

Normalisation is another important technique that standardises the data across the collected dataset. Indeed, in the context of music metadata, two composer name entries on two different Wikipedia pages could be referring to the same individual, however due to discrepancies in spelling, author preferences or even language disparities they could both have been entered into their pages in completely different spellings.

Creating algorithms to accurately map names that refer to the same artist to a single ID in a dataset is an essential portion of the data cleaning process [Javed et al., 2021, Eliyas and Ranjana, 2022]. This uniformity is essential for the functionality of content based recommendation systems which will heavily rely on the accuracy of this data for serving recommendations.

2.3.3 Data Preparation for Machine Learning Applications

Performing data encoding transforms the cleaned, categorical data into a numerical format that can be easily processed and utilised by recommendation algorithms. This is an important requirement for machine learning models which will require the data to be transformed into a numerical input.

Indeed, in the context of musical data, encoding techniques such as one-hot encoding convert the textual data of the composer and lyricist names into a numerical form that machine learning algorithm's can interpret, which will allow the dataset to be compatible with content based recommendation systems [Dasari and Varma, 2022, Bernardis et al., 2022].

2.4 Recommendation Systems

2.4.1 Fundamentals of Collaborative Filtering

Collaborative filtering is a cornerstone technique used by many popular music recommendation systems, including Spotify and YouTube. This method predicts a user's preferences based on the tastes and behaviours of a larger user community. Through analysing patterns of user interactions, such as song plays, likes and playlists, collaborative filtering algorithms can identify and recommend new music tracks that users with similar tastes have appreciated. This approach leverages the power of collective user activity to create highly personalised music discovery experiences [Moscati et al., 2023].

2.4.2 Challenges of Collaborative Filtering

Despite it's widespread use and success in powering recommendations, collaborative filtering faces notable challenges. One significant issue is the cold start problem, which occurs when new songs or artists are added to the platform. Without historical user interaction data, the algorithm struggles to recommend these new entries, potentially sidelining fresh and diverse content from reaching the listeners. Additionally, collaborative filtering can reinforce popularity bias, where popular tracks become increasingly recommended, overshadowing lesser-known music that might align with a user's preferences but lacks the same level of user engagement [Zhang et al., 2023].

These limitations highlight the need for complementary approaches that can address the gaps left by collaborative filtering. Content-based recommendation systems emerge as a promising solution, offering a different angle for personalising music discovery.

2.4.3 Fundamentals of Content-Based Recommendation Systems

Content-based recommendation systems allow applications to deliver personalised suggestions and completely transforms how end users can discover content. Through the analysis of the attributes of particular content, systems can create similar scores across content, to determine which content an end user would most likely enjoy next. This approach not only keeps users engaged with the content provided by the system, but also offers them an improved, tailored experience [Thorat et al., 2023].

2.4.4 Utilising Music Metadata for Recommendations

Content-based recommendation systems leverage content attributes to predict the sort of content a user would most likely enjoy next. In the context of music recommendations, a system could analyse musical metadata including the genre of the song, the composer and lyricist information to recommend other songs that share similarities with those a user has previously shown an interest in.

This method relies on the quality of the metadata and requires data collected through means of data scraping, to be thoroughly cleaned beforehand for the recommendations to be accurate. Without this, the systems will be unable to accurately match users with the content that they would most likely enjoy next [Javed et al., 2021].

2.4.5 Cosine Similarity Distribution

Once data has been properly cleaned, collected and encoded into a numerical representation, machine learning algorithms can use a cosine similarity to model similarity metrics between songs, determining how closely related one song's music metadata is to the next. Based on these similarity scores, we can determine the relationships of one song, to the top-n songs that are similar, to which we can serve as recommendations to the end user [Budiman and Giri, 2020].

2.4.6 Benefits of Content-Based Recommendation Systems

A significant advantage content-based recommendation systems have is that they do not heavily rely on needing a substantial amount of user data interactions to build personalised recommendations. This is especially important for creating engaging systems for new users who might have had little or no interaction with the system beforehand [Dasari and Varma, 2022].

Furthermore, since content-based recommendation systems are generated from the similarity analysis of a content's attributes, it is easier for a system to make it clear to their end users exactly why they were recommended that piece of content, which can further strengthen the trust the end user has to the system provider [Eliyas and Ranjana, 2022].

Chapter 3

Data Collection System

This chapter outlines the strategic approach and technical processes employed in constructing a comprehensive dataset of music metadata, primarily sourced from Wikipedia's vast discographies. The objective was to navigate through the platform's unstructured and varied data formats to extract accurate and complete metadata, with a focus on regions such as Iran, India and Turkey. I will discuss the methods I took to gather the discography pages and the various data collection techniques I implemented in my scripts. This chapter will also cover the various challenges I had to overcome throughout the process and detail the reasoning behind the various design decisions I made.

3.1 Strategy for Data Collection

Creating an efficient data collection system was a necessary component to building a comprehensive music dataset for Middle Eastern music. The main goal I devised was to construct large datasets from Wikipedia, a resource that is rich in music discographies that were in an unstructured and inconsistent format.

3.1.1 Initial Strategy and Challenges

I began analysing the structure of artist's individual Wikipedia pages to see if I could determine the necessary patterns I could use to design my scripts around. Wikipedia pages specifically separate their content through the use of headings, which i initially sought out to use in my scripts to be able to separate the content in order to isolate the data I would want to scrape for my dataset and get rid of the excess.

Typically an artist's page would contain at least the following sections:

- Title
- Introduction
- Early Life
- Career
- Discography
- Awards

Despite this overarching structure most pages followed, there was considerable variation in the headings used and even in their spelling (E.g. 'Songs' header instead of 'Discography'), introducing a level of complexity to extracting the specific content I was searching for. Indeed, the formatting of the content within these discography sections varied significantly, from bullet points to structured tables. My initial strategy was aimed at just extracting these discography sections to clean and organise the data into structured python data frames. However this approach revealed several challenges. The variability in section headings from page to page, coupled with the inconsistent details provided on artist pages, posed significant obstacles. Most artist pages opted for bullet points to list their works, adding further inconsistency. Often, the type of data contained within these bullet points wasn't explicitly indicated within the list itself but inferred from a nearby HTML element, often a type of header element, that would lead to significant inconsistency going from one page to the next.

3.1.2 Refining the Data Collection Approach

Faced with these challenges and the realisation that bullet points rarely contained the detailed metadata I sought due to their basic structure of mostly just containing the song name itself, I pivoted towards a more structured source. Artist discography pages, in contrast, offered a more organised layout with structured tables that included comprehensive metadata such as composers, lyricists, tuners and the songs associated with them. This discovery marked a pivotal shift in my data collection strategy, focusing on extracting data from these discography pages to ensure the richness and accuracy of the dataset.

I began with implementing a method of extracting just the pages that contained the discographies of artists specific to Iran, India and Turkey. This approach did pose a challenge of being able to collect sets of pages that contained just the discographies of artists that belonged to a particular country. There was a manual approach that involved going through the top artists of the country, visiting their Wikipedia pages manually and searching their page for a link to their associated discography page. This approach again introduced more variability in how the link to the discography page would be positioned in the page and similar spelling concerns, and was not a valid solution to the problem. Indeed, I did come up with a solution by using the record's feature in Wikipedia that displayed every single discography by an artist's nationality. From here, I could sort the pages by their required language, for instance for Iran I sorted the pages by Farsi, to which the record displayed all the pages written in Farsi that contained artist discographies. This was important as pages written in the language used predominantly by the country I was searching the data for, were more likely to contain accurate and complete metadata entries.

3.2 Developing Web Scraping Scripts in Python

3.2.1 Implementing the Scraper

Once I had collected all the necessary discography links, I began implementing a scraper, utilizing many Python libraries including Beautiful Soup which allowed me to parse the HTML pages, Requests to make HTTP requests to grab the HTML data, and Pandas DataFrame for data manipulation. These tools were necessary to navigate the structure of each Wikipedia page where this very structure varied from page to page.

Indeed, a common issue I had was the inconsistency in the format to which authors of each page had written the data in, where important information like song titles, composers, and lyricists could be scattered across different sections of the page or even presented in various formats including both bullet point and table formats. These Bullet point lists, a frequently chosen option to store lists of songs for any artist, often lacked comprehensive metadata and varied heavily in the HTML tags used from page to page [Tripathi and Nema, 2019]. For these factors I chose to omit scraping bullet point lists from the functionality of my scraper.

3.2.2 Addressing the Variability of Data Formatting

I did eventually come to the realization that the tables submitted by users, despite often being presented in various formats, were more likely to contain complete metadata, including the sought-after composer and lyricist information.

I devised a scraper that was capable of identifying and extracting every single table on a Wikipedia page and then filtering those tables to retain only those that contained music metadata (identified by column names containing words such as 'song name', 'composer', 'lyricist' for their respective language etc.) This not only solved the issue of data being scattered across different sections but could also automatically associate composer and lyricist information with individual songs through analyzing the tables row by row.

I extended my system to handle variations in languages across Wikipedia pages, whereby It can not only scrape and sort the tables that reference song name and album name to identify both song and album tables independently, but also create Pandas data frames from these tables and then further filter these data frames to only keep the columns that contain the music metadata I require, including the song name, composer, lyricist, and even tuning information – all for their respective language.

3.2.3 Final Data Pre-Processing

I further processed these raw data frames into structured CSV files. Using the panda's library, I processed the tables row by row to create three different files: songs.csv, albums_info.csv, and artist country.csv where each was designed to provide a standard structure to the dataset and begin to reveal the connections between the songs, albums, and artists, with each file having an individual ID to represent each respective record [Yuan, 2023]. Indeed, the artist.csv allowed me to build an artist specific dataset that would include all the artist names and their system generated artist ID's to which I could attach to each individual song record to create a link between the artists dataset and the songs dataset for each country, to structure my data properly.

Additionally, I also set up the publicly available Musicbrainz dataset. This involved setting up local instances of the MusicBrainz PostgreSQL server, and converting them into a similar songs, albums, and artist CSV format through appropriate SQL queries written through close attention to the documentation they provided on their website, which would allow me to provide a benchmark for comparison as to the quality of data I was scraping and pre-processing.

Chapter 4

Data Cleaning System

In this chapter we will discuss the meticulous process of transforming the raw, collected data from Wikipedia into a pristine, structured format that can serve as the backbone for a content-based recommendation system. This process is pivotal as the integrity and usability of content-based recommendation systems are directly hinged on the cleanliness and precision of the underlying data. We will delve into the multifaceted stages of data cleaning, each tailored to address specific challenges and intricacies inherent in the collected dataset.

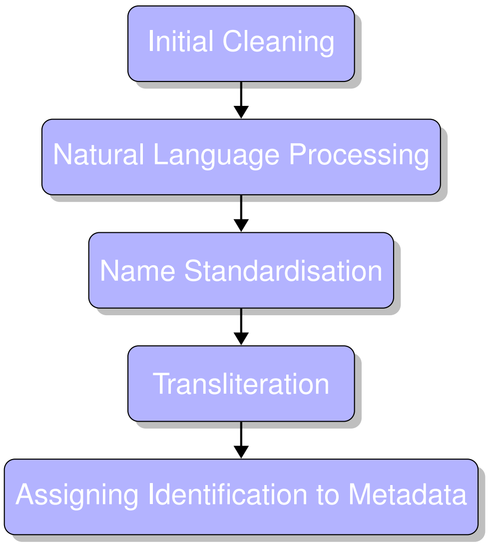

Figure 4.1: Data Cleaning Pipeline for Iran

4.1 Initial Cleaning Techniques

4.1.1 Basic Cleaning Steps

The data cleaning process required many crucial steps that were essential in transforming the collected raw data into a standardized, consistent, and usable dataset. This step was essential for ensuring our content-based recommendation system served accurate recommendations. The beginning phase of the process handled all forms of basic cleaning that were necessary.

This included:

- handling duplicate rows.
- handling incomplete entries.
- stripping leading and trailing white spaces.
- removing unnecessary symbols such as quotes and other language specific symbols.
- removing brackets and square brackets along with the text contained within them, as this information was often irrelevant and cluttered the dataset.

4.1.2 Data Unification

The second cleaning stage focused on addressing two primary issues: managing rows representing various distinct songs and consolidating multiple rows associated with the same song.

For song names encompassing multiple titles that were identified by commas or other indicators of a union between songs, such as 'and' or the '&' symbol, I divided these into individual row entries for each song title whilst preserving their original, associated metadata.

```
Songs Splitting:
Before Split: Abersar De Papad, Aaja Billo Katthe Nachiye,
    Beautiful Jatti
After Split: ['Ambersar De Papad', 'Aaja Billo Katthe Nachiye
    ', 'Beautiful Jatti']
```

Listing 4.1: Songs Splitting Example

For different rows that all represented the same song, but contained different metadata entries, I unified them into a single row, and aggregated the combined metadata into lists. For example, if two rows had the same song, but different composers, I would combine the composers into a list for just the single row. This not only reduced redundancy in my dataset, but also strengthened the dataset as single song entries could now store any number of composers, lyricists, and tuning names.

4.2 Using Natural Language Processing Techniques

The third cleaning stage utilized natural language processing techniques, POS tagging, and lemmatization, to further clean the data.

4.2.1 POS Tagging for Song Names

My system tokenized each word in a given name string, and used the POS tagging model to distinguish between nouns and non-nouns. Based on this, I developed a system that filtered out any song names that were both two tokens or more in length, and contained more than one non-noun phrase to remove data that did not represent the names of an artist.

4.2.2 Lemmatization of Album Descriptions

Additionally, lemmatization was implemented to further clean my description data I had gathered for my albums. This took each word in the description, and returned the base form of it, commonly known as the lemma. This normalized the descriptions to their lemma forms which ensured varying verb tenses and forms were standardized.

4.3 Standardising Names with Fuzzy Matching

4.3.1 Creating a Mapping System

The fourth cleaning stage was focused on standardizing and cleaning the names across entries, utilizing fuzzy matching. To solve the issue of songs referencing the same metadata name, but either spelling the name incorrectly, or simply using a variation of their name, I created a mapping system that took every name entry in the dataset, and used fuzzy matching to compare its similarity to every other name. The fuzzy matching process would then provide a score on how similar the two names were; from here the system decided that if any two strings had a score of 90 or above, which was very high, then they are referring to the same name and thus will add both names as keys to a dictionary mapping. Here I take the two names, and based on a rule of which artist name is longer, the system stores the longer name as the value for both keys. I decided to implement this rule favoring the longer name because, after already performing multiple cleaning processes, the longer name at this point in the pipeline typically represented the variation that contained the most data out of the two options.

4.3.2 Applying the Mapping to Enhance Dataset Consistency

After this process is repeated for every single name used in the composer, lyricist, tuning and artist columns, I use this mapping to replace every instance of every key in the dataset with its associated mapping value. This created an effective means to solve the variation problem associated with artist names, and also works across all languages that use Unicode and ASCII strings since fuzzy matching does not look at the meaning of the words but instead compares the similarity of the strings themselves. I further extend this process to handle songs and album rows from the same artist that have similarly high fuzzy scores, and combine their collected metadata into a single row to represent that piece of music.

```
Before Combining (Score:100):
Artist ID: 4
Row 1985: Song: Kuchi Kuchi, Composer: 741, Lyricist: 1372, Tuning:
Row 2824: Song: Kuchi Kuchi, Composer: 741, Lyricist: , Tuning:

After Combining:
Combined Row: Song: Kuch Kuch, Composer: 741, Lyricist: 1372, Tuning:
```

Listing 4.2: Example of Fuzzy Matching for Songs

4.4 Transliteration of Iran Datasets

The final stage of data cleaning transliterated the Iranian dataset, which mapped each character from the Farsi text stored in the dataset, into English. This was a crucial step in presenting the dataset in a format that was readable for users primarily using the Latin alphabet without relying on a translation method that could ultimately make mistakes and translate names incorrectly.

4.5 Assigning Identification to Metadata

I incorporated each of the composers, lyricists and tuning artists, discovered from the previous steps, into the main artist dataset. Each added artist was assigned their own uniquely generated artist ID, which was then linked back to their corresponding songs in the main song file. This not only expanded my artist dataset to capture a wider range of artist roles but also provided a more accurate view of the musical ecosystem.

4.6 Automation of the System's Pipeline

To ensure the adaptability of the project, I developed a bash script that would automate all of the scraping and cleaning processes for the datasets. This script was designed with scalability in mind, currently supporting three major country datasets: Iran, India, and Turkey. I chose to not implement the transliteration functionality into my cleaning pipeline due to it significantly increasing the processing time of my system, taking several hours to complete due to its heavy workload.

This helped achieve reproducible results and a seamless method to update the datasets to the most accurate and current versions, keeping in mind the Wikipedia pages that sourced the data are constantly being updated with new information all the time.

Chapter 5

Content-Based Recommendation System

In this chapter, I explore the development and implementation of a content-based recommendation system, a tool that transforms the way users discover music. By utilising my collected and cleaned datasets, this system leverages advanced machine learning and natural language processing techniques to offer personalised music recommendations by analysing the unique metadata of each song, including composer, lyricist and tuning information. The system creates a nuanced understanding of musical compositions, enabling the system to match users with songs that resonate with their existing musical preferences.

5.1 Metadata Processing

5.1.1 System Architecture

I developed a content-based recommendation system that utilized the cleaned datasets and was equipped with both machine learning and natural language processing techniques. This system was designed to create personalized music recommendations based on the metadata I had gathered.

The system takes a song name as input and queries the dataset to gather its associated metadata, including composer, lyricist, and tuning information for each song entry.

5.1.2 One-Hot Encoding

From this, the system encodes the entire dataset using one-hot encoding. For each song, the system encapsulates their lists of composers, lyricists, and tuning information into a unique identifier for each song named creative ids.

Here, one-hot encoding translates these creative ids into a binary format. This is essential as it allows the computation of cosine similarity scores to be generated across the dataset, which allows the system to identify songs that are similar to each other based on their creative ids.

Table 5.1: Original Dataset Entries for Selected Songs

Song Name	Composer Name	Lyricist Name
Dil Deewana	Sajid Wajid	Shabbir Ahmed
Mere Jeevan Saathi	Nadeem Shravan	Shameer

Table 5.2: One-Hot Encoded Dataset for "Dil Deewana" and "Mere Jeevan Saathi"

Song Name	Sajid Wajid	Nadeem Shravan	Shabbir Ahmed	Shameer
Dil Deewana	1	0	1	0
Mere Jeevan Saathi	0	1	0	1

5.2 Similarity Computation and Recommendations

The generated similarity matrix maps the top-n songs that have the highest similarity scores and returns them as the output along with their associated composer, lyricist and tuning information offering transparency to the end user into how the system calculated their recommendations. This system ensures each recommendation is not only personalized based on the song the user enjoys, but also that the recommendations are rooted in the metadata context of the song [Kumar et al., 2023].

Table 5.3: Top 10 Recommendation Results for "Dil Deewana" [Composer: Sajid Wajid, Lyricist: Shabbir Ahmed, Tuning: Unknown]

Song	Composer	Lyricist	Tuning	Similarity Score
God Tussi Great Ho	Sajid Wajid	Shabbir Ahmed	Unknown	100%
Fariyad	Sajid Wajid	Shabbir Ahmed	Unknown	100%
Ishqan Da Ishqan Da	Sajid Wajid	Shabbir Ahmed	Unknown	100%
Deewane Dil Ko Jaane Na	Sajid Wajid	Shabbir Ahmed	Unknown	100%
Dupatta Tera Nau Rang Da	Sajid Wajid	Shabbir Ahmed	Unknown	100%
Mastana Albela	Sajid Wajid	Unknown	Unknown	70.71%
Vande Matram	Shabbir Ahmed	Shabbir Ahmed	Unknown	70.71%
Ye Main Kahan	Sajid Wajid	Unknown	Unknown	70.71%
Dhoondte Reh Jaaoge	Sajid Wajid	Unknown	Unknown	70.71%
Area Ka Hero	Sajid Wajid	Jalees Sherwani	Unknown	50%

Chapter 6

Web Platform

In the pursuit of creating a sophisticated web application to utilise the datasets I had created, this chapter delves into the intricate architecture of a full-stack application that meticulously divides responsibility between backend and frontend systems. This division of tasks between the two subsystems ensured each could evolve independently to accommodate growing demands and new functionalities.

6.1 Backend Development using Django

The backend of my project is a Rest server built using Django, a Python web framework, as a means of serving API data requests. This transforms my datasets into accessible, structured JSON data that any application interface could utilize.

6.1.1 Data Management

Through implementing the backend to handle my extensive music database, I organised the collected and cleaned data into the following models with their associated attributes, that leverage Django's powerful Object-Relational Mapping capabilities to define and manipulate different database entities for efficient data retrieval.

- **Artist Model:** artist_name, link, aliases, country
- **Song Model:** song_name, album_id, artist_name, artist_id,composer_name, composer_id, lyricist_name, lyricist_id, tuning_name, tuning_id, description, country
- **Album Model:** album_name, album_id, artist_name, artist_id, description, country
- **Song Report Model:** song_name, issue

To effectively expose my models to my front-end application, several API views were implemented. These components were crucial for creating RESTful web services that would allow for the many operations my application would require.

The following services have been implemented for my API to provide access to:

- **Artist, Song and Album Access:** The system offers comprehensive access to detailed information about artists, songs and albums. This includes all the metadata I have collected and cleaned.
- **Country Specific Filtering:** Ability to filter songs, artists and albums based on the country.
- **Pagination for Scalability:** To address the challenge of handling vast datasets and to minimise the computational burden on my frontend, my backend breaks down the data delivery into manageable chunks. This significantly improves load times and resource efficiency.
- **Real-Time Search Suggestions:** An integral part of the user experience is the provision of search suggestions, which dynamically update as the user types.
- **User Report Handling:** This inclusion of a reporting feature for songs underscores the importance of maintaining accurate data for the application of a reliable content based recommendation system.
- **Country Statistics:** Calculates the total artists, songs and albums for a selected country.
- **Personalised Song Recommendations:** This view works in combination with the country specific filtering and the python scripts I wrote to perform one-hot encoding on the data and apply cosine similarity methods to sift through the relevant datasets to find songs that are similar to the user's chosen song, based on the metadata I have collected and cleaned.
- **Analyse Artist Contributions:** This feature offers in depth analysis of an artist's contributions for both other collaborators and for the industry. This view works in combination with the country specific filtering and the python scripts I wrote to analyze the dataset to build networks of connections that represent the artist's interactions with other artists.

6.1.2 Security Features

Django's network configuration enabled me to manage the access permissions to the API, which further ensured our system was accessible exclusively to the front end host and secured against unauthorized attempts to access it. Security was also ensured through the Django admin page, which served as an endpoint only a user with admin login details could access whereby they could manage and alter the data stored in the SQL database. I also mitigated cross-site scripting attacks by configuring the cors-header module. This feature allowed only the resources requested from my frontend to be accepted, blocking unauthorized hosts.

6.2 Frontend Development using ReactJS

6.2.1 Component Reusability and Efficiency

The frontend of my project was written using the ReactJS framework. I chose to develop the system using React over plain JavaScript, to create fast, interactive react components that I would be able to reuse over multiple pages. This allowed me to reduce duplicate and redundant code and create a streamlined and high-performance frontend application.

I also leveraged React's hooks to give components the ability to manage variable logic under various circumstances. This included the useEffect hook which allowed the system to re-render components visually based on side effects such as users typing in search bars or users interacting with the application and the useState hook which allowed my components to directly manage state variables such as data acquired from my backend.

6.2.2 Bootstrap Implementation

To further enhance the user interface design, I integrated the Bootstrap React framework, which leveraged its library for stylish components and animations to create a visually appealing and highly interactive web application that could display the data fetched from the backend.

6.3 Full Stack Architecture and Cloud Integration

6.3.1 Separation of Concerns

Splitting my system into independent backend and frontend systems was a deliberate choice as it allowed me to separate the concerns of each system to ensure they could independently scale on their own, without affecting the other. This allowed me to maximize performance and made maintaining each individual system a lot more straightforward, whilst at the cost of increased complexity overall [Villamizar et al., 2016].

Indeed, as the dataset expands or the number of users accessing the application increases, the backend's resources could be scaled to handle the extra traffic without affecting the frontend.

6.3.2 Digital Ocean Integration

I hosted both components on Digital Ocean's cloud infrastructure, which provided a flexible means to control the resources allocated to each system [Beigi-Mohammadi et al., 2020].

6.4 Data Insights

Providing data insights was an important feature within our music recommendation system, as it allowed us to delve deeper into the collected datasets and understand the links stretching across them. I split this feature into two major components: collaboration insights and industry insights.

6.4.1 Collaboration Data Insights

Collaboration insights for an artist show the detailed statistics of how many other musicians and creators contributed to that artist's music. For instance, the artist collaboration insights for a composer would show all the artists who performed that composer's songs.

These insights not only display to the end user the different collaborations an artist has had, but by ranking the collaborators from highest to lowest based on their total number of collaborations with each other, we quantify each relationship and can show the artist's most frequent collaborators.

My system's implementation involves analyzing the dataset and aggregating all instances of an artist's work, then parsing each participant's contribution to their individual role. The ranking system generated is also utilized by the bar chart visualization on my frontend system, that provides a visually pleasing and digestible means of consuming the data, allowing the end user to easily identify common patterns in the data [Holm, 2012].

6.4.2 Industry Data Insights

Industry insights for an artist extend the previous collaboration insights by mapping out an artist's indirect connections within an industry, showing professionals with whom the artist is connected through mutual collaborations. My system tracks primary but also secondary and tertiary relationships to accomplish this feat. For instance, if a composer composed a song for a singer, this section would show all the direct collaborators along with all the other composers who have worked with that singer, mapping the indirect links from that composer to the other composers.

6.4.3 Interactive Data Visualisations

To further enhance the user experience, I used multiple react libraries to create interactive visualizations of these data insights on my front end. I created both an interactive bar graph and a node network graph, where both allowed users to explore the data under different perspectives and to see for themselves all the different connections linked between the artists. Indeed, the interactivity of the graphs is central to the user experience and allows the user to hover over elements to reveal additional information, and to click on nodes and bars to dive deeper into that artist's individual data insights [Chi et al., 2016].

6.5 User Interface Design and User Experience Journey

6.5.1 Navigation

Upon launching, the web application introduces the users to a home menu, which is anchored along with the other main pages of the application to a navigation bar at the top of the page. This allows users to seamlessly transition between the different sections. Leveraging React Router's capabilities, the site employs route based component loading, dynamically setting the current content based on the URL.

This approach enhances the user experience by enabling smooth page transitions without requiring entire page reloads, maintaining a fluid interaction throughout the user's journey and significantly reducing load times.

6.5.2 Home Page and Country Discovery

The home page serves as the entry point to my application, displaying an 'About The Project' section that outlines the methodologies I employed to create my datasets, and three dynamically generated card components. Each card represents a different country, and provides a snapshot of the latest statistics drawn from my API.

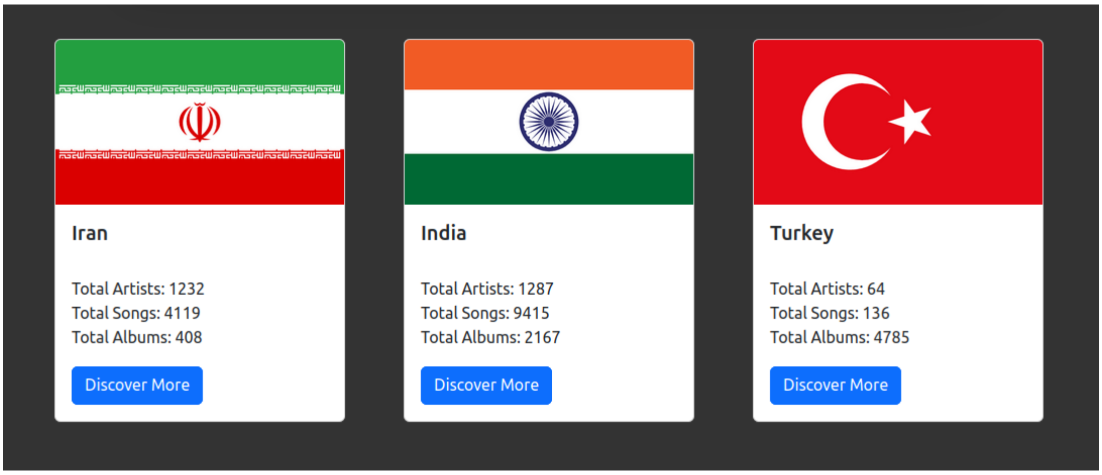

Figure 6.1: Home Page: CountryCards.

6.5.3 Artist Exploration

The cards each include their own 'Discover More' button, which can navigate the user to the artist's dataset tab, and automatically adjust the country select field on this page to match the chosen country, triggering the application to fetch the first paginated batch of artists for this country from the API. The resulting artist cards are then constructed to offer a visual representation of each artist.

The integration of a search bar on this page allows users to seamlessly navigate through the large dataset of artists. After the input of the second letter in the search bar, subsequent characters will instantaneously prompt a query to my API, fetching the top suggested names that align with the entered text, which will pop up as clickable suggestions that can automatically fill the rest of the search query. This feature not only enhances the search bar's functionality but also encourages user engagement by providing immediate, more relevant feedback as the search term evolves.

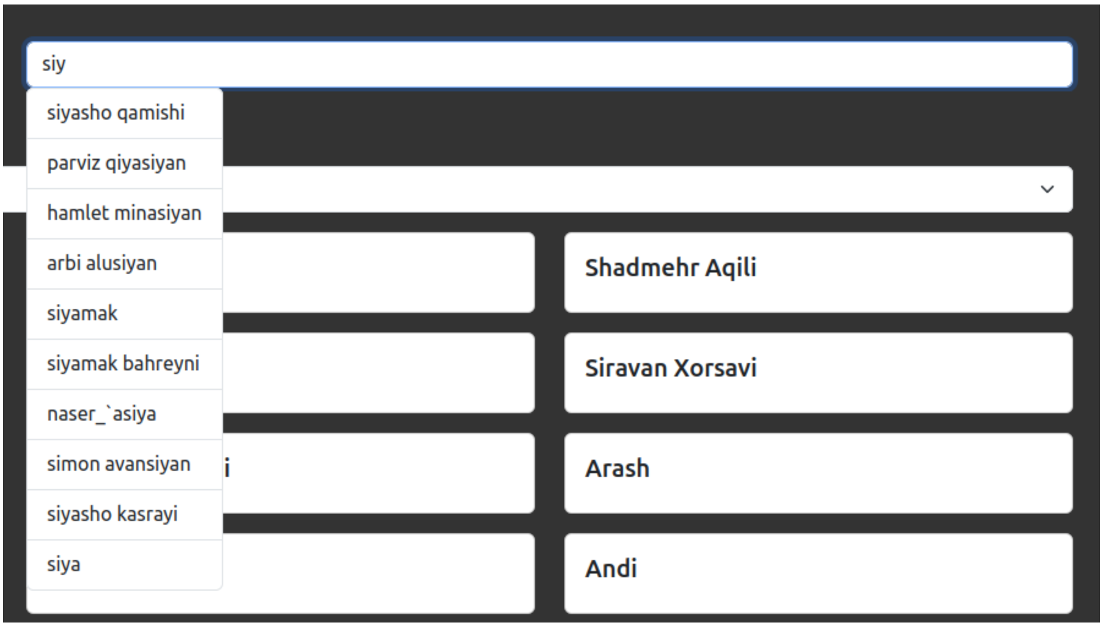

Figure 6.2: Artists Dataset Page: Suggestions Feature.

6.5.4 Artist Data Insights

Selecting an artist card will transition the user to a detailed data insights tab that is dedicated to the chosen artist. Once this page begins to load, the application performs a series of intricate requests to my API that are designed to fetch information about the artist. This includes a comprehensive list of songs performed, composed, written or tuned by the artist alongside details about each song's collaborators, and their list of albums performed.

The frontend will construct card components for all the songs and albums, and dynamically load these cards in the form of paginated lists. This section also includes both a songs and albums button, where depending on which one is selected by the user, will only render that list of paginated cards to the user.

6.5.5 Dynamic Insight Generation and Visualisations

Simultaneously, the API will calculate the collaborative dynamics among composers, lyricists, tuning artists and performers, ranking these collaborators based on the frequency of their interactions with the artist. The frontend will then dynamically assess the availability of the data from these API requests to construct buttons for accessing the collaborative insights. Selecting these buttons will not only update the paginated list of collaborators but also toggle between the visualisation graph rendered next to the paginated list.

This graph component is by default displayed as a bar chart but also includes a button that will render a node network graph instead. Both these graphs visually represent the intricate web of collaborations where each node and bar, representing artists and their frequency, serve as an interactive element. Hovering over these elements will reveal further details and clicking on them can redirect users to their respective artist's data insights page, enhancing navigational fluidity. Since the majority of the page's components will have their state computed dynamically based on the artist the user has selected, this page is not accessible as its own tab in the navbar, and is instead meant to be accessible through clicking an artist's name anywhere on my application.

Figure 6.3: Artist Data Insights Page for Siyasho Qamishi: Insights and Visualisations.

6.5.6 Song Card Functionalities

Each song card on the page is designed to manage the state of its associated metadata, where the presence of this data is indicated by an orange downward emoji. This emoji tells the user that they can click on this song card to expand it to reveal further information. Once clicked, in addition to rendering the associated composer, lyricist and tuning collaborators for the song, I have provided two critical buttons: 'Report Issue' and 'Discover More' to introduce further layers of user engagement.

- **Report Issue:** This button will render a modal that allows users to report discrepancies or issues related to the song. The submitted report will be encapsulated by the associated ID of the selected song and the user's inputted feedback and posted to my API for storage, further underlying my commitment to data accuracy and its importance in building effective content based recommendation systems.

- **Discover More:** This button ties into the content-based recommendation system, and once clicked will navigate the user to the Top-N Recommender tab. This page includes a search engine optimised for uncovering similar songs. This feature will initialise the song name into the search bar of this page and automatically begin the searching process, which will prompt my backend to calculate and return the top 20 recommendations based on their similarity scores and associated artist metadata.

6.5.7 Content Based Recommendation Queries

Given the computational intensity of generating recommendations, the website manages loading states of the API requests, displaying animations to keep users informed during the processing times. This ensures a seamless user experience even when complex back-end calculations are taking place.

The recommendations page features a search bar implementing a suggestions feature, that is activated after the entry of two letters in the search bar. Similar to the artists search bar, this will generate clickable suggestions of songs in the database that match the inputted text, that can automatically fill the rest of the search bar once clicked. This was designed to make interacting with my content based recommendation system as seamless as possible.

6.5.8 Paginated Lists for User Exploration

The remaining pages of my application include both the songs and albums dataset pages, which are designed to showcase paginated lists to allow users to explore my datasets without overwhelming them with their sheer size. Each page organises the content into individual cards which represent songs or albums and are aligned with the country selected through a dedicated select component on each page.

On the songs page, each song card is clickable which will unveil linked data for the performers, composers, lyricists, and tuning artists associated with the song. Each of these roles is presented as clickable elements, which will navigate the user to the respective artist's data insight page. Each song card also incorporates a 'Discover More' and 'Report Issue' Button, mirroring the artist's data insight page functionality. Whilst album cards are also clickable, currently their functionality is limited to navigating to their performing artist's data insights page.

Chapter 7

Experimental Results

In this section I will analyse the findings and outcomes derived from the evaluation of my data cleaning system, focusing on my Wikipedia sourced dataset. I will be assessing the impact of my methodology on enhancing the metadata quality of the Wikipedia data, an important aspect for the development of an accurate and reliable content-based recommendation system. The analysis will begin highlighting the contrast between the dataset's states before and after the application of the cleaning procedures.

Subsequently, I will extend the analysis to cover the distribution of metadata across the songs within the final datasets, focusing on both Wikipedia and Musicbrainz. This examination will facilitate an understanding of both the structure of the data and the prevalence of metadata, offering an insight into the dataset's usability for a content-based recommendation system.

7.1 Dataset Quality Analysis: Before and After Cleaning

In this section I analyse the changes in the quantity of unique lyricist and composer data within our datasets for Iran and India, before and after the data cleaning process takes place. The quality of the lyricist and composer data is of vital importance to the dataset's usability in a content-based recommendation system. Indeed, during the design of the data cleaning process, a pivotal aspect was to be able to handle multiple variations of metadata names to ensure each artist was represented correctly in the dataset.

Through this investigation of the total number of unique lyricists and composers before and after the cleaning process, I aim to:

- Evaluate the effectiveness of my data cleaning methodologies in refining the dataset through both my mapping system utilising fuzzy matching and general duplicate and inaccuracy removal functions.
- Highlight the importance of meticulous data cleaning processes in the context of building comprehensive and usable music metadata datasets, which are important for creating accurate and reliable content-based recommendation systems.

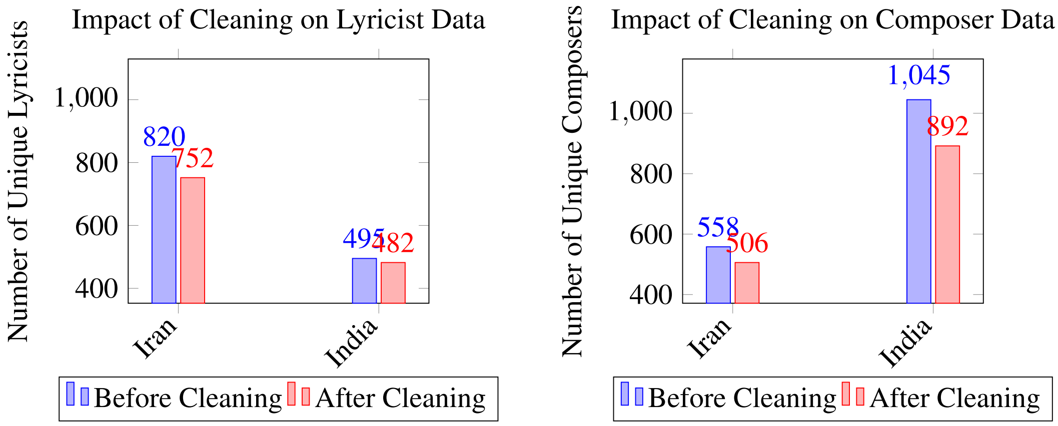

Figure 7.1: Impact of data cleaning on music metadata for lyricists and composers.

The figures visualise the impact of the data cleaning process on the number of unique composers and lyricists for both the Iran and India portions of my Wikipedia datasets.

The data revealed a 9.3% decrease in the total number of composers for Iran, and a substantial 14.6% decrease in the total number of unique composers for India, both suggesting a significant portion of the original data was effectively identified and cleaned by the various methods implemented in my data cleaning pipeline.

Similarly, the analysis extended to lyricists, where the data cleaning process also resulted in a decrease in the number of unique lyricists. For Iran, there was an 8.3% reduction in the count of unique lyricists and for India, a modest 2.6% decrease in the number of unique lyricists. Whilst these changes were smaller compared to the composer data, they still reflect a significant impact of the cleaning process in refining the quality of my dataset.

7.2 Metadata Distribution Analysis: Wikipedia vs. Musicbrainz

Following the analysis of the impact of the data cleaning process on the number of unique composers and lyricists in our Wikipedia datasets for Iran and India, I will extend my examination to compare the distribution of the songs with at least one instance of composer or lyricist information against the songs without any metadata for each dataset. This comparison is made between my collected and cleaned Wikipedia dataset, and the data obtained from Musicbrainz for the same countries. This analysis aims to:

- Demonstrate the contrast in the richness of the metadata between the Wikipedia and Musicbrainz Dataset, highlighting the effectiveness of Wikipedia as a source for detailed composer and lyricist metadata.
- Emphasize the significance of possessing a high count of unique metadata entries in the context of developing an accurate and reliable content-based recommendation system.

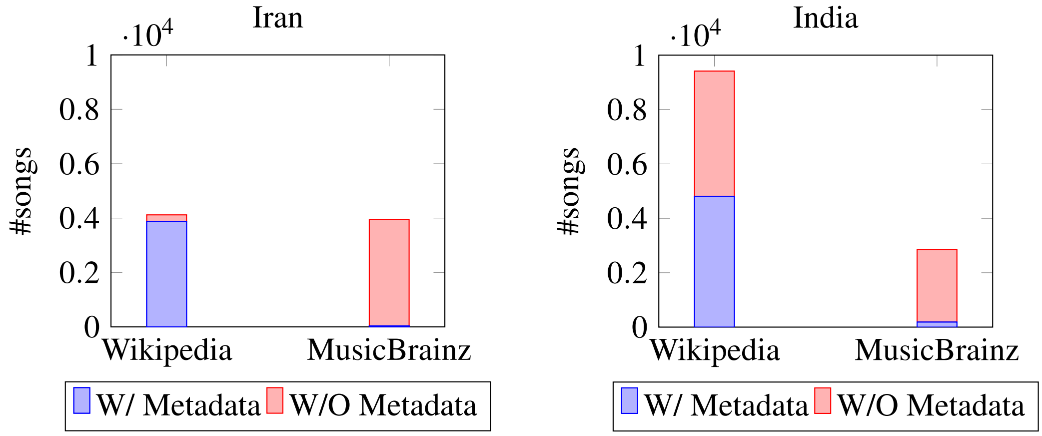

Figure 7.2: Comparative analysis of Metadata in Both Datasets for Iran and India. These figures illustrate the distribution of songs with at least one instance of Composer/Lyricist Metadata versus songs without, for both Wikipedia and MusicBrainz.

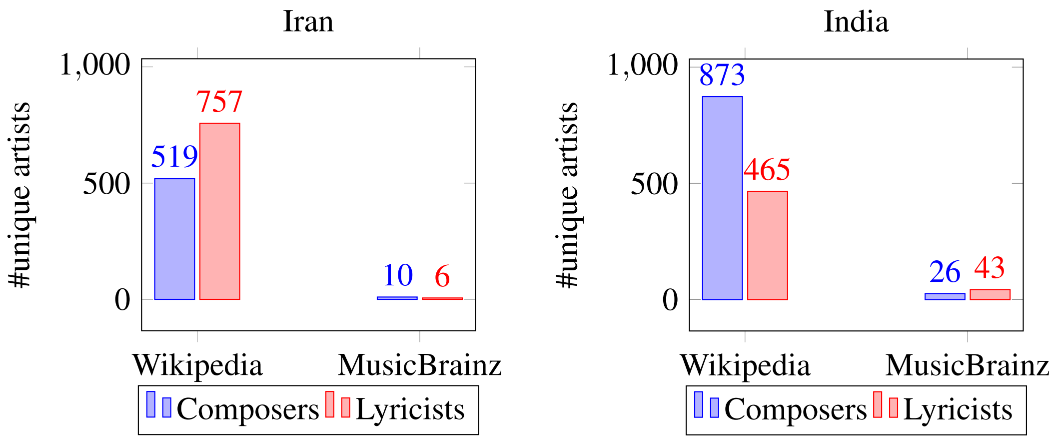

Figure 7.3: These figures illustrate the number of unique artists for each metadata role for both Wikipedia and Musicbrainz.

The analysis reveals significant differences in not only the distribution of metadata, but also in the number of unique composers and lyricists across the Wikipedia and Musicbrainz datasets for both Iran and India. Wikipedia's dataset not only showcases a significantly higher proportion of songs with unique metadata, defining its richness compared to Musicbrainz, but also illustrates a larger quantity of unique composer and lyricist information. For Iran, the disparity was especially pronounced, where Wikipedia provided nearly 94% songs with metadata compared to Musicbrainz 1.2%. Similarly, India's Wikipedia Dataset contained a substantial 51.1% of songs with metadata, vastly outperforming Musicbrainz's 6.5%. Further exploring the unique contributors within these datasets, Iran's Wikipedia results for unique Composers and Lyricists were overwhelming. The Wikipedia dataset contained a catalog of composers approximately 51.9 times larger than that of MusicBrainz, and it's list of lyricists was approximately 126.2 times greater. The results were mirrored for India, whereby India's Wikipedia dataset for composers was roughly 33.6 times larger than MusicBrainz, and for lyricists, it was approximately 10.8 times greater.

7.3 Summary of Results

7.3.1 Impact of Data Cleaning on Metadata Quality

The data cleaning process led to a noticeable reduction in the total number of unique composers and lyricists for both Iran and India, signifying the removal of inaccuracies, duplicates and irrelevant entries. Despite the reductions being more substantial for composers than for lyricists, they both reflected a significant enhancement in the dataset's quality, making it more streamlined and relevant for application in a content based recommendation system.

7.3.2 Comparative Analysis with Musicbrainz

When examining the distribution of songs with metadata against the songs without metadata for each dataset, a significant contrast emerged between the Wikipedia and Musicbrainz datasets. The Wikipedia dataset for Iran displayed nearly 94 times the proportion of songs with metadata compared to Musicbrainz's 1.2%. For India, the different was also remarkable, with Wikipedia containing almost 8 times the proportion of songs with metadata than Musicbrainz 6.5%. These results highlight Wikipedia's superior metadata richness and underscores the specific value designing my methodologies around this source of data bring.

Delving deeper into the comparison of unique contributors further displayed Wikipedia's dominance. In Iran, Wikipedia's entries for composers and lyricists were approximately 51.9 and 126.2 times larger, respectively, than those documented in MusicBrainz. Similarly in India, Wikipedia's documentation of composers and lyricists was about 33.6 and 10.8 times more extensive, respectively, than MusicBrainz's. This disparity between the two datasets further illustrated Wikipedia's potential as a source for researching musical metadata, particularly for the development of content based recommendation systems.

7.3.3 Implications for Content-Based Recommendation Systems

My analysis of the experimental results allowed me to derive the following interpretations:

Enhancement of Metadata Accuracy Through Data Cleaning: The data cleaning process which led to a notable reduction in the total number of unique composers and lyricists highlighted the critical role of data quality focused methods in the context of a content based recommendation system. By effectively removing the inaccuracies, duplicates and irrelevant entries, the cleaning process not only streamlined the dataset but also significantly improved its accuracy and relevance. Indeed this approach to data cleaning, specifically through the use of a name mapping system utilising fuzzy mapping, ensured that the remaining metadata more accurately represented the true contributions of each artist, laying a solid foundation for the recommendation system to make more accurate and meaningful recommendations.

Superior Metadata Richness of Wikipedia Data The comparative analysis clearly demonstrated that data collected from Wikipedia boasted a higher degree of metadata richness compared to that of MusicBrainz. This enhanced richness was evidenced by the greater proportions of unique metadata entries in the Wikipedia dataset for both Iran and India. This difference not only highlighted Wikipedia's capability to provide more detailed and varied metadata but also signified the potential for Wikipedia-sourced data to provide more detailed and rich metadata as a new source of data to be utilised in the construction of content based recommendation systems.

The study emphasises the critical role of data quality, particularly metadata richness and accuracy in the design of a content based recommendation system. The findings advocate for the quality of my methodology's ability to construct a dataset with metadata quality previously unparalleled.

Chapter 8

Conclusion

In the conclusion of this research, I summarise the significant advancements made in enhancing music metadata for Middle Eastern countries. Additionally, I will also explore the challenges and limitations I faced during the development of my project. Following this, I will discuss the implications of my research and outline plans to expand this project in the future.

8.1 Summary of Findings

8.1.1 Breakthrough in Music Metadata for the Middle East

My research project marks a major advancement in the field of music technology, particularly in regards to the gap found in comprehensive music metadata for Middle Eastern countries.

Through successfully creating a system capable of building music datasets with extensive composer and lyricist metadata, and through comparing this to the current datasets available, I have shown the impact my system has on the current landscape and the potential for it to scale even further.

Indeed, the reproducible nature of my data collection process should serve well in the continuation of my project, as it's independent nature will allow support for more and more countries to be implemented.

8.1.2 Development of a Full Stack System

Being able to develop a full-stack system that leveraged the datasets allowed me to provide an easy and intuitive way for users to access not only the raw data itself, but also personalised recommendations and calculated data analytics.

My system displays the integration of not only advanced data scraping and cleaning techniques, but also advanced recommendation and data insight algorithms. These capabilities allowed my system to serve personalised recommendations to each user, based on the comprehensive metadata I collected, and detailed data insights that allow a user to explore the data in a meaningful and enjoyable way, providing the user with a deeper understand of each artist and their works.

8.1.3 Superiority of Wikipedia Datasets

Indeed, the datasets built from this research were found to contain significantly more comprehensive metadata than the existing MusicBrainz database, which not only helped bridge the gaps found previously in middle eastern metadata, but served to provide a way for me to calculate and serve meaningful recommendations for this dataset, which through composer and lyricist information would not have previously been possible for the MusicBrainz dataset.

8.1.4 Reproducibility and Scalability of Research Methods

The methods I utilised in my project, including the collection and cleaning of the music metadata, were created in a reproducible and scalable means that would allow my research to be easily maintained and extended.

The use of bash scripts to control the collection and cleaning python pipelines not only ensured my methods were reproducible and intuitive to run for a user, but also as a means for my system to update the databases and maintain the latest data at all times, accounting for Wikipedia pages being updated over time to record more valuable composer and lyricist information. This approach is essential to ensure my research stays relevant over time [Varin, 2015].

8.2 Limitations of the Project

Despise the results achieved, it's important to acknowledge the limitations encountered the construction of my methodology. These limitations will highlight the challenges faced and provide valuable insight for future work in the field of music metadata analysis.

8.2.1 Integration Challenges with Discogs Dataset

Other than Musicbrainz, I initially set out to further enrich my dataset collection with the Discogs public database, which would have provided another point of references for me to compare the efficacy of my methodology. However, the formatting of the Discogs data dumps in XML posed significant challenges during the data collection phase. Moreover, unlike MusicBrainz who initialised user setup through a docker containerised instance of a PostgreSQL server, specifically designed to handle computationally expensive queries to access the data, The XML format of the Discogs dumps posed many difficulties in maneuvering through the XML data to extract the relevant composer and lyricist metadata efficiently, especially considering the storage capacity of the data dumps. This limitation underscored the importance of considering data format and accessibility when making large datasets publicly available.

8.2.2 Challenges in Gathering Comprehensive Discographies

The main limiting factor encountered in the project was the process of gathering comprehensive discography pages from Wikipedia. Although a method was developed to acquire these discographies, expanding the dataset to include a broader range of discographies would likely require manual searching, or more efficiently a method to crowd source contributions from the public. This limitation is significant as the scope and depth of the dataset will directly influence the performance of my recommendation system, and the potential future of scaling my project. However, it should be noted my current system of attaining Wikipedia discographies nets the following total number of discographies, sorted by country.

8.2.3 Reliance on Tabular Data

The project's methodology heavily relied on data stored in tabular format, as this was deemed the most reliable way for my system to ensure the quality of the data scraped. This was a strategic choice made to mitigate the risks associated with the quality of the data found in less structured formats. By concentrating on tabular data, the approach inherently bypassed some information available in more free form formats. However, this decision was crucial in ensuring the integrity and reliability of the data collected, laying a solid foundation for developing an accurate and dependable recommendation system.

8.3 Implications of Research

8.3.1 Contributing to Middle Eastern Music Datasets

My research has many implications on the current state of music technology, that extend past the enhancing of music meta data for the middle eastern countries. By bridging this gap in the metadata, specifically for Iran and India, this project did not only enrich their datasets with comprehensive metadata but established a robust and reproducible means to extend these efforts to other underrepresented regions.

8.3.2 Potential for Global Contributions

Indeed, this successful implementation of a recommendation and data insights system through the reproducible and scalable means of collecting and cleaning detailed music metadata, proves the utility this data has and the potential to establish this data for other parts of the world.

These methods not only serve as a means to develop detailed datasets but also to ensure that the songs themselves reflect the combined efforts of the artists who created those songs [Hardjono et al., 2019].

8.4 Future Work

8.4.1 Plans for Global expansion

Whilst I succeeded in the initial goals I set out to achieve at the beginning of my project, I envision a future for this project that will expand its reach on a global scale. Indeed not only do I plan to refine the data cleaning systems and pipelines I have put in place, but also to significantly scale the system to support a larger array of countries.

Whilst this would not be an easy task, my efforts into making the system reproducible and scalable should help in making expanding the data coverage as smooth a process as possible, and eventually support a global reach.

8.4.2 Community Driven Expansion for Language Support

The main limiting factor in my system is finding the individual key words in languages specific to each country, that are required for my system to support scraping the Wikipedia pages written in those websites.

I plan to expand my system to support contributions from a global community, whereby the project could rapidly accumulate knowledge on the various keywords that signify the data my scraper needs to target when faced with languages it doesn't currently support. This model if implemented correctly would accelerate the expansion and further increase the accuracy of my system.

8.4.3 Enhancements to Full Stack System

As well as scaling the data collection and cleaning systems, I also have a vision of substantially enhancing the full stack system that utilises the datasets. Indeed, I plan to extend the capabilities of my API to handle more queries and increased traffic and performance, which will accommodate the dataset as it grows alongside it.

I plan to expand the frontend to increase the number of features available and overall make it a more intuitive and enjoyable experience for the user to use. This not only involves adding more ways to dive into the data but also to add more personalisation features along with the recommendations to give the user an enhanced experience.

Due to the separation of concerns structure I have implemented, isolating the backend logic from the frontend, I also plan to develop more frontend technologies such as an app or a chrome extension, that could independently scale alongside the other technologies I have already developed.

8.4.4 Extending System to handle Lyrical data

Beyond composer and lyricist metadata, I plan to explore collecting the lyrics of songs. This inclusion of lyrics would open more means for analysis and could give my system even more means of determining a profile that describes the themes and emotional aspects of a song. This layer could have an extremely beneficial impact on the personalisation of my recommendations in combination with the composer and lyricist metadata I have already gathered.

8.4.5 Commitment to Preserving Culture

Whilst this road map lays many ambitious and difficult tasks, I aim for this future work to not only solidify my project's standing at the forefront of musical technology, but to also reinforce its commitment to preserving the rich tapestry of global musical heritage.

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Appendix A

First appendix

Table A.1: Impact of Data Cleaning on Music Metadata

Country	Unique Lyricists		Unique Composers	
	Before Cleaning	**After Cleaning**	**Before Cleaning**	**After Cleaning**
Iran	820	752	558	506
India	495	482	1045	892

Table A.2: Comparative Analysis of Metadata Rows for Iran and India

Dataset	Iran (#songs)		India (#songs)	
	W/ Metadata	**W/O Metadata**	**W/ Metadata**	**W/O Metadata**
Wikipedia	3871	248	4808	4607
MusicBrainz	35	3920	185	2673

Table A.3: Number of Unique Artists for Each Metadata Role

Country	Dataset	Composers	Lyricists
Iran	Wikipedia	519	757
Iran	MusicBrainz	10	6
India	Wikipedia	873	465
India	MusicBrainz	26	43